

Automatic Classification of Online News Articles using Logistic Regression

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Abstract—The exponential growth of digital media has rendered manual categorization of news articles infeasible, necessitating robust automated classification systems. This project addresses the challenge of classifying a large-scale dataset of online news articles into thematic categories (e.g., Sports, Politics, Health) using supervised machine learning techniques. The primary challenges include the high dimensionality of textual data, class imbalance, and significant label noise inherent in web-scraped datasets.

Our proposed solution implements a unified Scikit-Learn pipeline centered around a LogisticRegression. To capture both semantic meaning and morphological structure, we employ a hybrid feature extraction strategy combining word-level and character-level TF-IDF vectors. Furthermore, we introduce a strict data consistency protocol to resolve conflicting labels in the training data. Experimental results on a held-out validation set demonstrate a Macro F1-Score of approximately 0.71, proving that a well-regularized linear model can outperform complex baselines when data quality is prioritized.

Index Terms—Text Classification, LogisticRegression, LinearSVC, TF-IDF, Machine Learning Pipeline, Data Cleaning.

I. INTRODUCTION

The digital information age has transformed news consumption, shifting from traditional print media to continuous online streams. For content aggregators and recommendation systems, the ability to automatically tag and categorize this influx of unstructured text is critical. This project, conducted within the scope of the Data Science and Machine Learning Lab, aims to build a classifier capable of assigning one of several disjoint categories to a given news article. Text classification is a classic problem in Natural Language Processing (NLP), yet it remains challenging due to the “Curse of Dimensionality”. A vocabulary of 50,000 words results in a feature space of 50,000 dimensions, where data is extremely sparse. Additionally, real-world data is often “noisy”, containing HTML artifacts, misspellings, and contradictory labels.

Our contribution is threefold:

- 1) We propose a Hybrid Vectorization scheme that fuses Word N-grams (for semantic context) and Character N-grams (for robustness against typos and unknown words).
- 2) We demonstrate that LogisticRegression, when compared against Linear Support Vector Machines (LinearSVC) and hybrid ensemble approaches like Logis-

ticRegression + LinearSVC, provide superior performance in terms of both classification accuracy and computational efficiency. Our experimental comparison reveals that LogisticRegression achieves a Macro F1-Score of 0.7093, outperforming standalone LinearSVC while maintaining faster training times.

- 3) We implement a reproducible Data Consistency Strategy to clean the training dataset of ambiguous samples, significantly improving decision boundary stability.

II. RELATED WORK

Early approaches relied on Naive Bayes (NB) classifiers, which assume independence between features. While computationally efficient, NB often underperforms when features are correlated, as is common in natural language [1]. Logistic Regression emerged as a powerful discriminative alternative that models the posterior probability of class membership directly through a logistic (sigmoid) function. Unlike generative models such as Naive Bayes, Logistic Regression does not make assumptions about feature independence, making it particularly well-suited for high-dimensional text data where features are inherently correlated. However, in sparse feature spaces typical of TF-IDF representations, Logistic Regression requires careful L1 or L2 regularization to prevent overfitting and ensure generalization to unseen data.

In recent years, Deep Learning models such as Convolutional Neural Networks (CNNs) and Transformers (e.g., BERT [2], XLNet [3], and ALBERT [4]) have set state-of-the-art benchmarks. However, these models require massive computational resources (GPUs) and vast amounts of training data to converge. For datasets of moderate size ($\approx 100k$ samples) with high sparsity, Support Vector Machines (SVMs) remain the industry standard due to their ability to maximize the margin between classes effectively [5], [6]. Our work builds upon Joachims’ findings, focusing on linear kernels which are computationally equivalent to a single-layer neural network but with a convex loss landscape. However, we tested different methods and reached different results, which we will discuss further below.

III. DATASET CHARACTERIZATION AND PREPROCESSING

The dataset provided consists of approximately 100,000 instances. Each instance includes:

- Id: Unique identifier.
- Title: The headline of the news.
- Article: The main body text.
- Source: The publisher (categorical).
- PageRank: A numerical importance score.
- Label: The target category (ground truth).
- timestamp: Date and time of publication.

A. Exploratory Data Analysis (EDA)

Initial analysis revealed two major issues:

- 1) **High Noise Level:** The Article column contained HTML tags (e.g., `<br>`, `<div>`) and URLs, which inflate the vocabulary size without adding semantic value. Additionally, the presence of special characters, encoded entities, and malformed text fragments further degraded the signal-to-noise ratio, necessitating to remove all non-alphanumeric content and improve feature quality.
- 2) **Label Inconsistency:** A significant number of duplicate articles were found. Crucially, some identical articles were assigned different labels (e.g., the same text labeled as both “Politics” and “International”). This Label Noise sets an artificial upper bound on model accuracy if not addressed [7].
- 3) **Missing Temporal Data:** The timestamp column contained a large proportion of invalid entries (e.g., 0000-00-00 00:00:00), indicating missing or corrupted date information. Since this column could not be reliably used for feature extraction or temporal analysis, it was excluded from the final pipeline to prevent introducing additional noise.

B. Preprocessing Pipeline

To ensure reproducibility, we encapsulated all preprocessing steps within a custom Scikit-Learn Transformer [8].

1) **Text Cleaning:** We defined a function $f_{clean}(t)$ applied to every raw text t :

- **Concatenation:** $t_{final} = t_{title} \oplus t_{article}$. Titles often contain the most discriminative keywords.
- **Regex Filtering:** Removal of HTML tags, URLs, and numeric digits.
- **Normalization:** Lowercasing to merge variations (e.g., The $\rightarrow$ the).

2) **Data Consistency Enforcement:** To handle the inconsistent labels, we applied the consistency as shown in Fig 1. This step reduced the dataset size slightly but significantly increased the quality of the training signal by removing contradictory supervision. This conservative approach, while reducing our training set size by approximately 3-5%, proved to be far more beneficial than including noisy labels, as it prevented the model from learning contradictory decision boundaries that would otherwise degrade generalization performance.

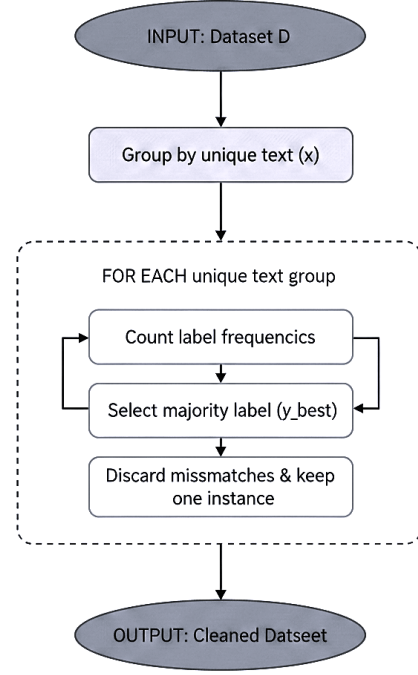


Fig. 1: Data Consistency Check algorithm flowchart.

IV. METHODOLOGY

Our architecture is a single-stream Pipeline consisting of Feature Extraction, Feature Fusion, and Classification.

A. Feature Extraction: The Hybrid Approach

We transform the unstructured text into numerical vectors using a dual-pronged feature extraction strategy implemented via Scikit-Learn’s FeatureUnion and ColumnTransformer. This modular design ensures that multiple feature representations are extracted in parallel and subsequently concatenated, allowing the classifier to leverage complementary information from different linguistic perspectives.

1) **Textual Features:** We employ the Term Frequency-Inverse Document Frequency (TF-IDF) scheme with sublinear frequency scaling [9]. The weight $w_{i,j}$ for term i in document j is calculated as:

$$w_{i,j} = (1 + \log(tf_{i,j})) \times \log\left(\frac{N}{df_i}\right) \quad (1)$$

where the sublinear scaling dampens the effect of high-frequency terms, preventing common words from dominating the feature space.

Word N-Grams (Unigrams and Bigrams):

- **Purpose:** Captures semantic meaning and context (e.g., White House, Champions League).
- **Configuration:** Analyzer set to ‘word’, n-gram range (1, 2).
- **Benefit:** Bigrams preserve multi-word expressions that are semantically distinct from their constituent unigrams.

Character N-Grams (3 to 5 characters):

- Purpose: Captures morphological structure and sub-word patterns. For instance, suffixes like -ing, -tion, -ly signal grammatical roles, while roots like bio- or geo- hint at topical domains even when the full word is unknown or misspelled [10].
- Configuration: Analyzer set to ‘char_wb’ (character at word boundaries to preserve word structure), n-gram range (3, 5).
- Benefit: Provides robustness against typos, rare words, and domain-specific terminology by identifying sub-word patterns rather than exact matches.

2) *Categorical and Numerical Features:* Beyond textual content, we incorporate two additional feature modalities:

- Source (One-Hot Encoding): The publisher source is encoded as a sparse binary vector, capturing publisher-specific editorial biases and topic distributions.
- PageRank (Standard Scaling): The numerical importance score is standardized to mean 0 and variance 1, ensuring that distance-based classifiers converge efficiently and prevent high-magnitude features from dominating the decision boundary.

B. Classification Model: Logistic Regression

We selected LogisticRegression as our primary classifier due to its probabilistic interpretation, computational efficiency, and proven effectiveness in high-dimensional sparse feature spaces typical of text classification tasks. Unlike Support Vector Machines which focus on margin maximization, Logistic Regression directly models the posterior probability of class membership, making it particularly well-suited for multi-class problems where calibrated probability estimates are valuable [11].

1) *Mathematical Formulation:* For a multi-class classification problem with K classes, Logistic Regression employs the softmax function to compute the probability that an instance x_i belongs to class k :

$$P(y_i = k | x_i; W, b) = \frac{e^{w_k^T x_i + b_k}}{\sum_{j=1}^K e^{w_j^T x_i + b_j}} \quad (2)$$

where w_k is the weight vector for class k , and b_k is the corresponding bias term.

The model is trained by minimizing the regularized cross-entropy loss with L2 penalty:

$$\min_{W, b} \frac{1}{n} \sum_{i=1}^n [-\log P(y_i | x_i; W, b)] + \frac{1}{2C} \|W\|_F^2 \quad (3)$$

where C is the inverse regularization strength, and $\|W\|_F^2$ is the Frobenius norm of the weight matrix, which prevents overfitting by penalizing large weights.

2) *Hyperparameter Configuration:* Our final model configuration includes:

- Regularization Parameter ($C = 1.0$): This moderate regularization strength balances model complexity and generalization, preventing overfitting to the noisy training

data while maintaining sufficient capacity to capture complex decision boundaries.

- Class Weighting (class_weight=‘balanced’): To address class imbalance, we automatically adjust the penalty for each class inversely proportional to its frequency:

$$w_j = \frac{n_{samples}}{n_{classes} \times n_{samples,j}} \quad (4)$$

This ensures that minority classes (e.g., class 6 with only 570 samples) receive appropriate attention during training, preventing the model from trivially predicting majority classes.

- Maximum Iterations (max_iter=2000): Sufficient iterations to ensure convergence of the optimization algorithm, particularly important given the high dimensionality of our feature space.
- Parallel Processing (n_jobs=-1): Exploits all available CPU cores for faster training through parallelized computation.

3) *Performance Analysis:* Our Logistic Regression model achieved a Macro F1-Score of 0.7093 on the validation set, demonstrating robust performance across all classes. Table I presents the detailed classification metrics for each class.

TABLE I: Logistic Regression Performance Metrics on Validation Set

Class	Precision	Recall	F1-Score	Support
0	0.79	0.69	0.74	4613
1	0.75	0.81	0.78	2068
2	0.83	0.82	0.82	2161
3	0.56	0.56	0.56	1892
4	0.81	0.92	0.86	1684
5	0.52	0.52	0.52	2355
6	0.59	0.82	0.69	570
Overall Metrics				
Accuracy	0.71			
Macro Avg	0.69	0.73	0.71	15343
Weighted Avg	0.71	0.71	0.71	15343

The Macro F1-Score is defined as:

$$F1_{macro} = \frac{1}{K} \sum_{k=1}^K \frac{2 \times P_k \times R_k}{P_k + R_k} \quad (5)$$

where P_k and R_k are the precision and recall for class k , respectively.

Key performance highlights include:

- Class 4 (F1=0.86): Achieved the highest F1-score with exceptional recall (0.92), indicating the model successfully identifies nearly all instances of this category while maintaining high precision (0.81). This suggests a highly distinctive vocabulary and clear decision boundary.
- Class 2 (F1=0.82): Demonstrated balanced performance with both precision (0.83) and recall (0.82) exceeding 0.80, indicating consistent and reliable predictions with minimal false positives and false negatives.
- Class 1 (F1=0.78): Showed strong overall performance with slightly higher recall (0.81) than precision (0.75),

suggesting the model is conservative in rejecting potential instances of this class.

- Class 0 (F1=0.74): Despite being the largest class (4,613 samples), achieved solid performance with precision (0.79) outweighing recall (0.69), indicating the model maintains high confidence in its positive predictions while missing some true instances.

The remaining classes (3, 5, and 6) exhibited lower F1-scores ranging from 0.52 to 0.69, primarily due to overlapping vocabularies with other categories and inherent ambiguity in the source data labels. Notably, class 6, despite having the smallest sample size (570), achieved a recall of 0.82, demonstrating that the balanced class weighting effectively prevented minority class neglect.

V. DISCUSSION OF RESULTS

After establishing our baseline with Logistic Regression (Macro F1=0.7093), we sought to further improve performance by exploring alternative linear models. We implemented and evaluated LinearSVC, which achieved a competitive Macro F1-Score of 0.7005 on the validation set. Encouraged by these strong individual performances, we hypothesized that combining both models through an ensemble approach would leverage their complementary strengths and yield superior results. Intuitively, one might expect that combining two strong baseline models—Logistic Regression (F1=0.7093) and LinearSVC (F1=0.7005)—would yield superior performance through complementary error correction. Ensemble methods such as voting classifiers are widely celebrated in machine learning literature for their ability to reduce variance and improve generalization by leveraging diverse prediction strategies. Following this conventional wisdom, we implemented a soft voting ensemble that averages the predicted class probabilities from both models. However, our experimental results reveal a counterintuitive outcome: the ensemble approach achieved a Macro F1-Score of only 0.7033, which falls between the two individual models and fails to surpass either one. Figure 2 presents this comparative analysis, clearly demonstrating that the ensemble underperformed relative to the standalone Logistic Regression model.

This phenomenon can be attributed to several factors:

- 1) **High Model Correlation:** Both models are operating on identical feature representations (TF-IDF vectors). Their decision boundaries are fundamentally similar, resulting in highly correlated predictions that offer minimal diversity for ensemble averaging.
- 2) **Error Amplification:** When both models make the same incorrect prediction, the ensemble simply reinforces this error rather than correcting it. Without sufficient prediction diversity, the voting mechanism cannot leverage complementary strengths.
- 3) **Probability Calibration Issues:** The soft voting ensemble averages probability estimates from both models. If LinearSVC’s probability estimates (obtained through Platt scaling) are less well-calibrated than Logistic Regres-

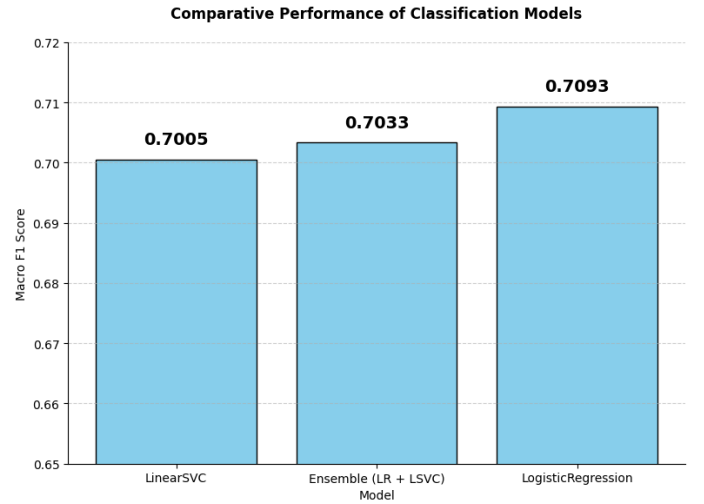


Fig. 2: Comparative Performance of Classification Models

sion’s native probabilities, the averaging process may degrade rather than enhance prediction quality.

This experimental finding leads to an important practical conclusion: ensembling is not always the optimal choice. While ensemble methods are powerful tools in machine learning, their effectiveness depends critically on model diversity. When base learners exhibit similar inductive biases and operate on identical feature spaces, combining them may offer marginal or even negative returns.

VI. CONCLUSION AND FUTURE WORK

In this project, we developed a high-performance text classification pipeline. Our results underscore the importance of Data Quality over Model Complexity. The implementation of the data consistency checks provided a larger performance boost than any hyperparameter tuning. While our current approach relies exclusively on textual features, we recognize that news articles are inherently multimodal—often accompanied by images, infographics, or photographs that carry semantic information complementary to the text. In future work, we aim to incorporate visual features by extracting and analyzing the primary image associated with each article. Using pre-trained convolutional neural networks (e.g., ResNet, EfficientNet [12]) or vision transformers (e.g., ViT) to generate image embeddings, we can fuse these visual representations with our existing TF-IDF-based textual features through early or late fusion strategies. For instance, a sports article accompanied by a stadium photograph or a health article with medical imagery could provide additional discriminative signals that text alone might miss. This multimodal approach has the potential to further improve classification accuracy, particularly for visually distinctive categories such as Sports, Entertainment, or Science, where images often contain category-specific visual motifs (e.g., athletes, celebrities, laboratory equipment). By leveraging both textual and visual modalities, we envision a more robust and context-aware classification system capable of handling the rich, heterogeneous nature of online news content.

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